Recitation: 11

Sample	Age	Income	Credit Score	Actual Class	Predicted Probability
S1	25	40	650	1	0.9
S2	30	60	700	1	0.75
S3	45	35	600	0	0.60
S4	50	30	580	0	0.40
S5	28	55	720	1	0.30

(1) Compute Confusion Matrix, 0.5 Threshold

(2) Metrics: Accuracy, Precision, Recall (TPR)
F1 Score

(3) FPR

(4) ROC Points

(5) TPR, FPR for 0.7, 0.6, 0.5, 0.4, 0.3

* Rule

If Probability $\geq$ Threshold = Predicted 1

If Probability $<$ Threshold = Predicted 0

S1	0.90	1	Class 1, Positive
S2	0.75	1	Class 0, Negative
S3	0.6	1	
S4	0.4	0	
S5	0.3	0	

1) Threshold value = 0.5

Prob $\geq$ = 1

Prob $\not\geq$ = 0

TP 2	FP 1
1 FN	1 TN

$$\text{Accuracy} = \frac{TP + TN}{\text{Total}} = \frac{2+1}{5} = \frac{3}{5} = 0.6$$

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{2}{3} = 0.67$$

$$\text{Recall (TPR)} = \frac{TP}{TP+FN} = \frac{2}{3} = 0.67$$

$$\text{F1 Score} = F_1 = \frac{2PR}{P+R} = 0.67$$

$$FPR = \frac{FP}{FP+TN} = \frac{1}{2} = 0.5$$

* Threshold = 0.7

$TPR = \frac{2}{3} = 0.67$, $FPR = 0$

* Threshold = 0.6

$TPR = 0.67$, $FPR = 0.5$

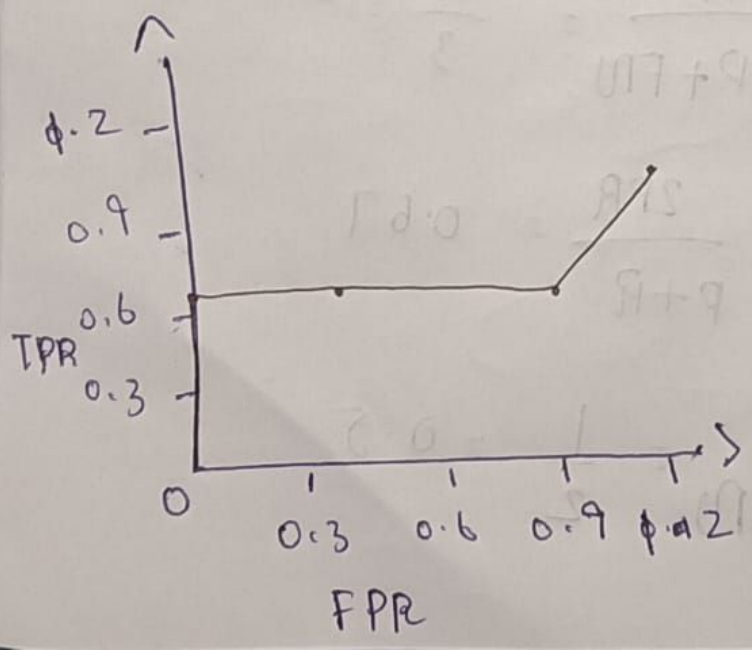
* Threshold = 0.4

$TPR = 0.67$, $FPR = 1$

* Threshold = 0.3

$TPR = 1$, $FPR = 1$

* ROC Points => (FPR, TPR)

=> (0, 0.67); (0.5, 0.67);

(1, 0.67); (1, 1)